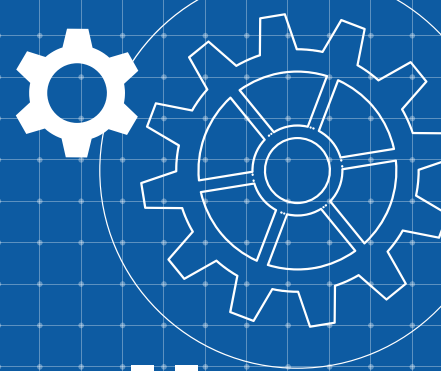




Hydrogen Fuel Cell Vehicle

Nora Jiang
Professor Carr Everbach

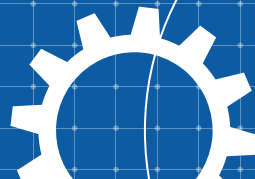




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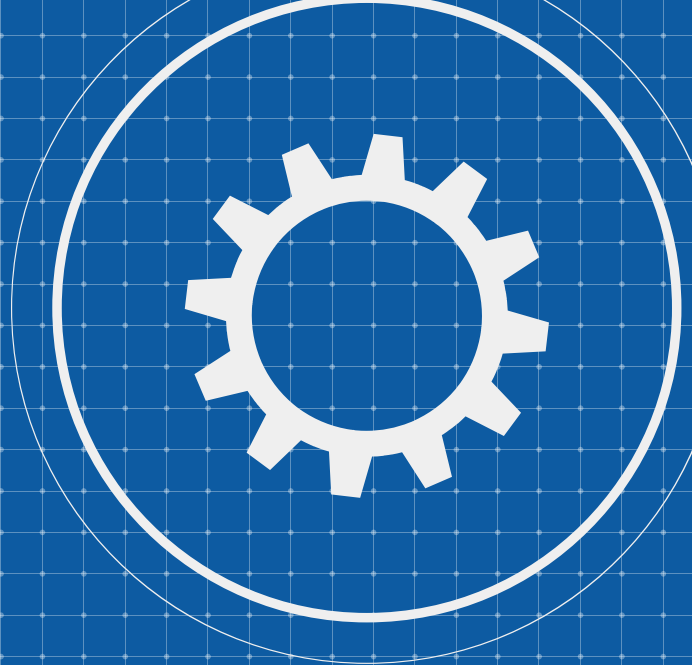




01

Introduction

What is a HFCV? And why is there one sitting in the High Bay?





What you need to know first



Hydrogen (H₂) as a fuel?

- H₂ is an energy carrier (like gasoline)
- Energy is produced through electrolysis
- Stored as a liquid before entering the fuel cell



Why switch to H₂

- H₂ is incredibly energy dense
- Clean in comparison to gas
- Large range before needing to refuel



Inside the Fuel Cell

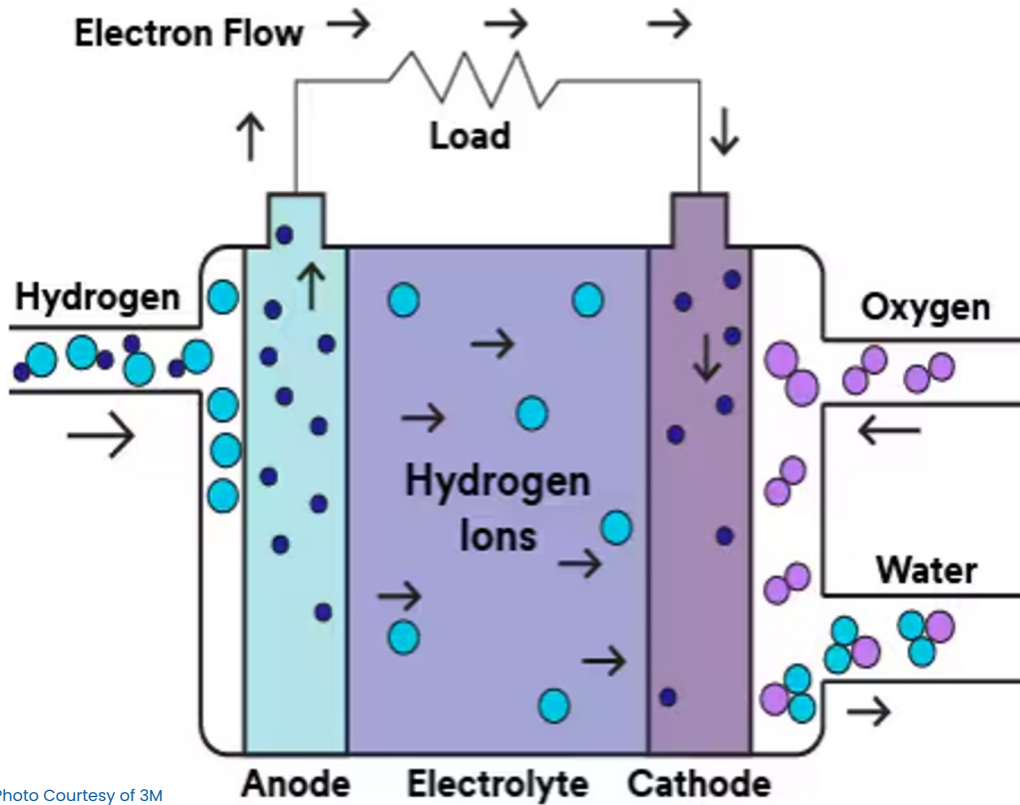


Photo Courtesy of 3M

Our Actual Fuel Cell



Shell Eco Marathon



What Is It?

Competition to make
the most energy
efficient vehicle

Why Compete?

Gives a target for our
work on the vehicle



Photo Courtesy of Shell Eco Marathon

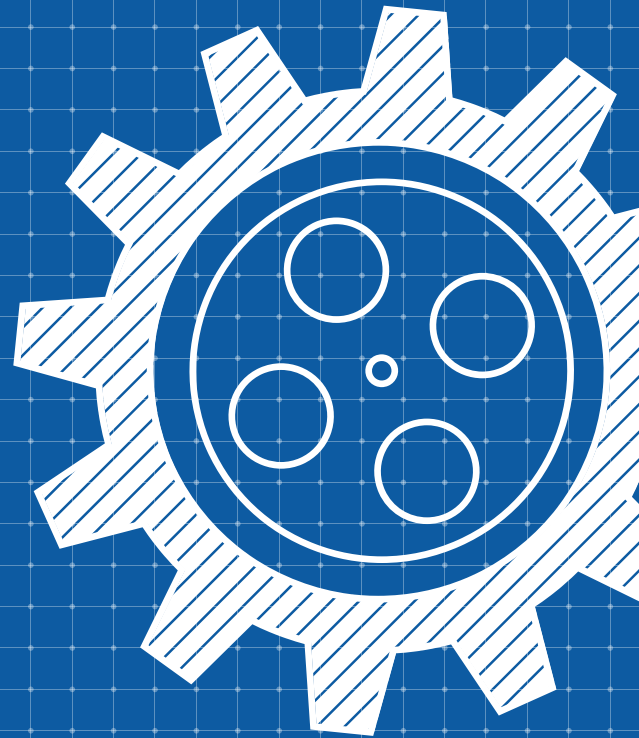




02

Current Project

Rebuilding the Chassis

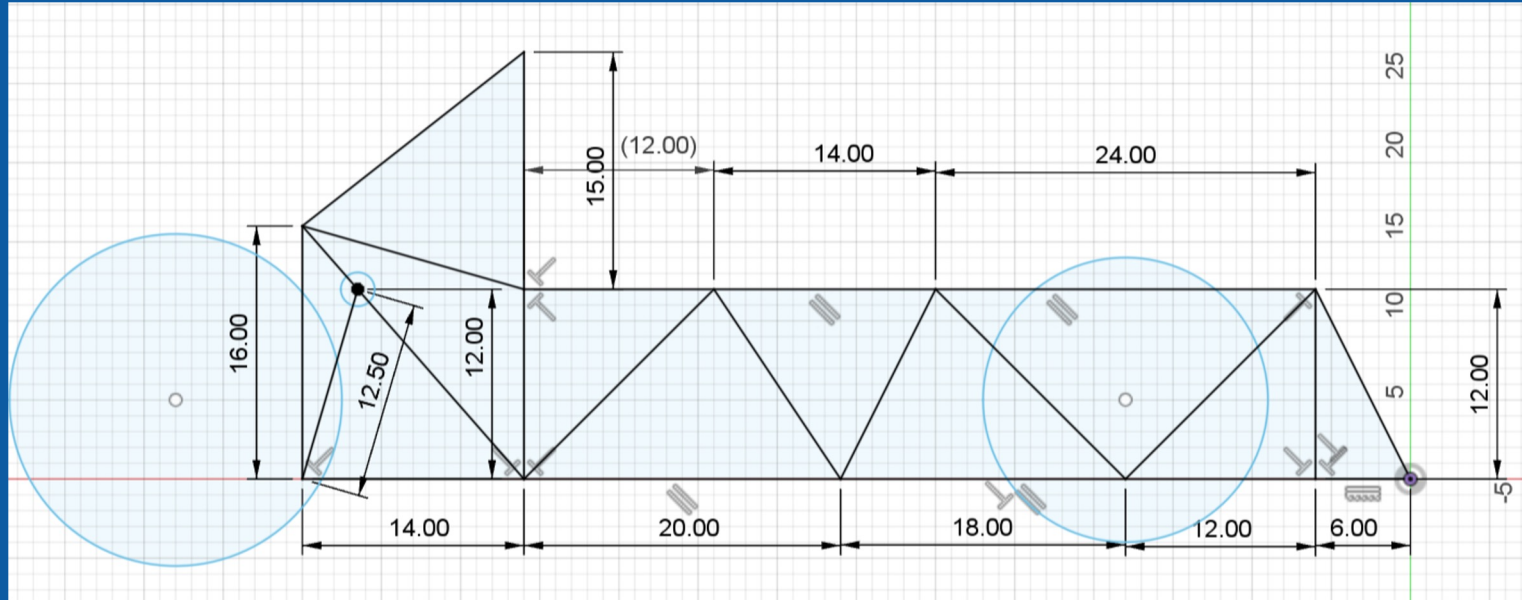


Before This Summer

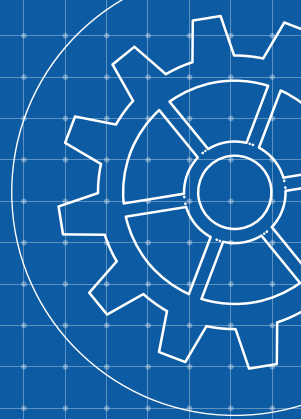
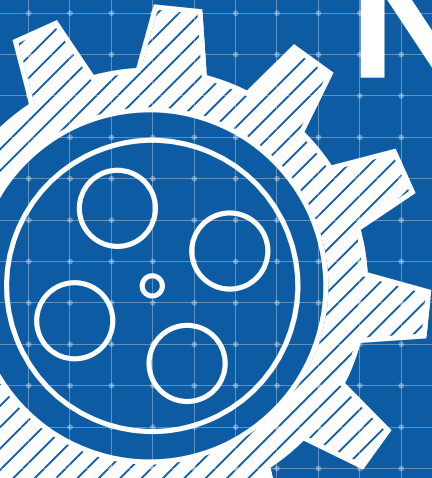
- Previous vehicle
 - No longer adheres to Shell's safety standards
 - Needs a complete rebuild before competition ready
- Revitalization w/ Ben, Jano, and Justin
 - Materials testing
 - Preliminary Chassis Design
 - Rear Brakes and Suspension system



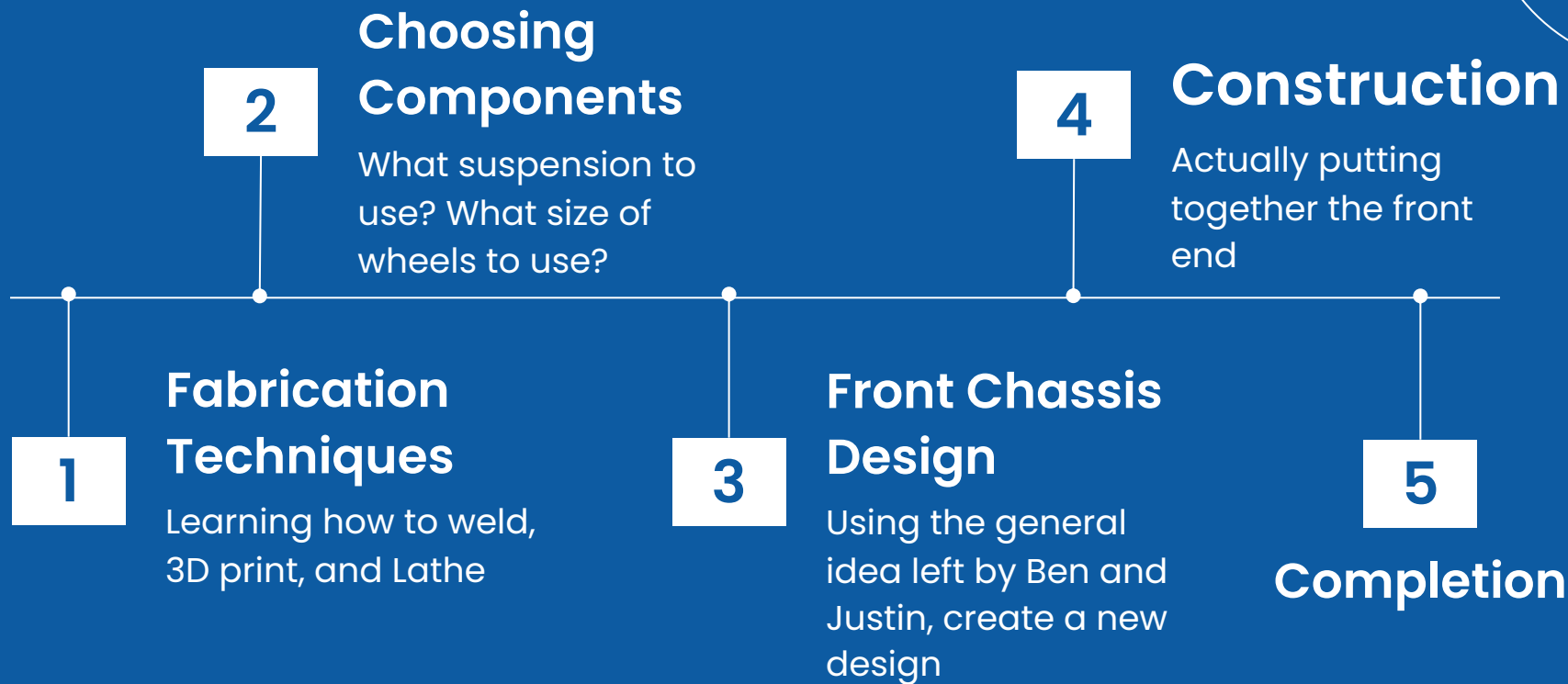
Rough Design From Ben and Justin



Goal: New Front End



What will that take?

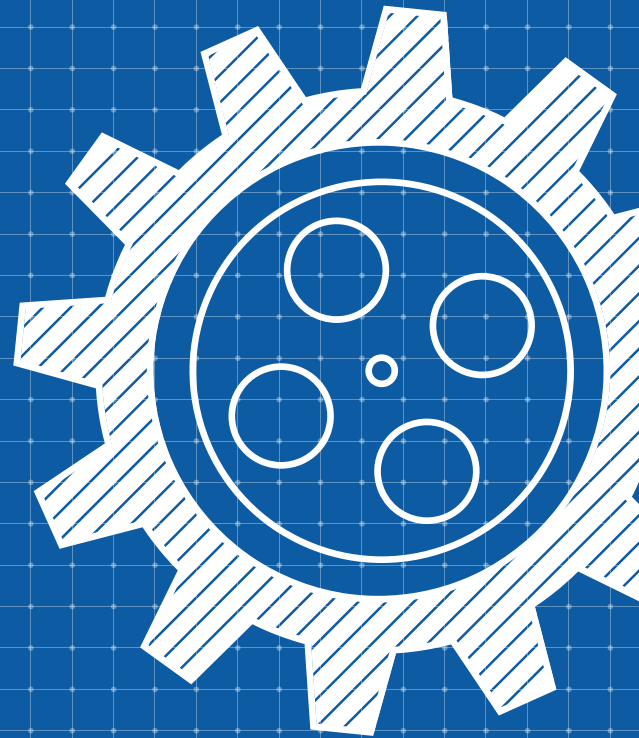


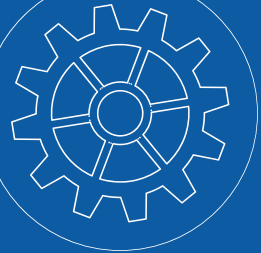


03

Future Considerations

Rebuilding the rest of the HFCV





Fall 2024



Rear of Chassis

Build and fit the
rear wheel +
suspension system



Floor of Vehicle

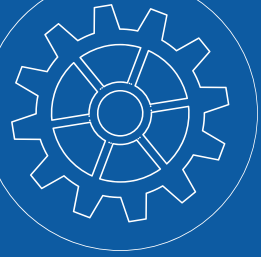
Purchase
components and
assemble



Electrical Components

Wire the new car
and put in the
hydrogen system





Spring 2025



Cowl Design and Build

Build the Cowl for
the vehicle so that
its not a skeleton



Compete

Transport Team
and Car to Indiana



Testing

Testing entire
system to the Shell
Eco Marathon
Requirements





Thank you for listening

Questions, Comments, Concerns?

njiang1@swarthmore.edu
avasquel@swarthmore.edu
ceverbal@swarthmore.edu

